



Secrecy, the patent puzzle and endogenous growth

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ABSTRACT

Endogenous growth models typically assume that innovations are protected solely by patents. Consequently, they predict that strengthening patent protection stimulates innovation by increasing the private return to R&D. This contrasts with empirical evidence that: (1) firms routinely employ a combination of patents and secrecy and (2) stronger patent regimes have been associated with increased patenting but not increased innovation, a finding referred to as the “patent puzzle.” In this paper, I develop a novel model of endogenous growth that is consistent with this evidence. Successful innovators’ choose an appropriation strategy in the form of a patent, secrecy mix. Both protection methods are imperfect, and innovators face a trade-off between secrecy’s superior protection against imitation and patenting’s protection against rival innovation. I demonstrate that growth oriented policy changes generate important economic effects through their impact on the endogenous appropriation strategies of innovators.

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1. Introduction

Endogenous growth theory has long emphasized the role of the patent system in driving economic growth. In traditional theory, patents preform an essential function; they create a private incentive to invest in R&D by allowing firms to appropriate the returns of their innovations through a period of monopoly profits. Since ideas are assumed to be public goods, firms have no incentive to conduct R&D without the legal right to exclude competitors from the use of an innovation. It follows directly that stronger patent protection increases the reward from innovating, thereby stimulating R&D investment and economic growth. However, empirical evidence does not support this simplistic view. The empirical literature consistently finds that the continued strengthening of the legal protection afforded to patent holders in developed countries has had no significant effect on innovation or productivity (Hall and Ziedonis, 2001; Branstetter, 2004; Bessen and Hunt, 2007). Summarizing the findings of twenty-three empirical studies, Boldrin and Levine (2009) state, “they find weak or no evidence that strengthening patent regimes increases innovation; they find strong evidence that strengthening the patent regime increases ... patenting!”

This counterfactual prediction of traditional endogenous growth theory, the so called “patent puzzle,” raises an important question: do firms rely on patenting to appropriate the returns to their innovations? Mounting evidence suggests that firms in most industries view patents as a relatively small component of their overall appropriation strategy (Cohen et al., 2002; Hall et al., 2014). First, firms routinely decide not to patent their innovations. Surveys of European and U.S. firms find that the average propensity to patent is between 30–55%.¹ Second, firms widely consider secrecy to be a more effective appropriation

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¹ Patent propensity refers to the proportion of innovations for which a patent application is made. See Cohen et al. (2002) and Hall et al. (2014) for details.

mechanism than patents (Arundel, 2001; Cohen et al., 2000; 2002; Hall et al., 2013). Patents offer only temporary protection and require disclosure of at least some of an innovation's underlying technical information. Even under strong patent rights, the disclosed information can be used by competitors in ways that either do not directly infringe on a patent, or that are difficult for the patent holder to detect. Although secrecy does not offer the same formal protection, it avoids disclosure and protects an invention indefinitely as long as it can be kept private.

When firms do patent, it is often for reasons *other* than protecting their innovation from imitation as typically assumed (Hall et al., 2014). In particular, patents are increasingly used strategically for their “blocking” effect on rival innovations (Shapiro, 2001; Cohen et al., 2002; Boldrin and Levine, 2013). Since modern innovation routinely requires building on many existing components and ideas, innovations typically overlap. Legal monopoly control over even seemingly minor aspects of a product or process can be used to extract rents from subsequent innovators through licensing agreements, or impede competing innovation directly through infringement litigation. As Boldrin and Levine (2013) argue, “the downstream blocking effect of existing monopoly grants on incentives for future innovation has greatly increased in recent decades because modern products are made up of so many different components.” In fact, firms often strategically apply for patents that cover only a small portion of their innovations (Anton et al., 2006). By selectively patenting some aspects of an innovation, firms gain protection against competing innovations while mitigating disclosure requirements and relying on secrecy to protect their most important intellectual property.

Indeed, more recent contributions to the endogenous growth literature have incorporated aspects of these strategic motivations for patenting. A series of papers including Chu (2009), Chu et al. (2012), and Yang (2018) model the consequences of overlapping patent rights as a required profit sharing between new and previous innovators. This profit sharing can inhibit entrants' incentives to conduct R&D, and generate a blocking effect on future innovation. In a similar vein, Dinopoulos and Syropoulos (2007), Şener (2008), and Davis and Şener (2012) investigate how firms use their patent rights to raise the cost of competitors' innovative efforts by investing resources in “rent protection activities” (RPAs), such as government lobbying and patent infringement litigation. While these papers highlight important effects of the patent system on economic growth, patents remain the sole appropriation mechanism by default. As a consequence, policy evaluations are based on the unrealistic assumption that firms always patent and rely fully on the patent system to create incentives to innovate. Necessarily, these papers omit the growth effects of policies through their impact on the appropriation strategies of firms.

On the other hand, the literature that has examined how firms use a combination of patents and secrecy in their appropriation strategies has typically been restricted to partial equilibrium settings (Anton and Yao, 2004; Denicolò and Franzoni, 2004; Kultti et al., 2007; Kwon, 2012). To my knowledge, Suzuki (2015) is the only study that has incorporated endogenous patenting and secrecy decisions in a general equilibrium growth context. Suzuki emphasizes that the impact patent protection has on R&D incentives depends upon the extent that firms utilize patents in their appropriation strategies. Consequently, he argues that strengthening patent protection may fail to stimulate economic growth if firms rely primarily on secrecy. While Suzuki's work illustrates that incorporating secrecy into an endogenous growth model can significantly alter its policy implications, he does not allow for patents to be used strategically to affect rival innovation. In this way, he omits not only a central motivation for patenting, but also a policy relevant channel by which firm appropriation strategies impact economic growth.

In this paper, I develop a novel model of endogenous growth that incorporates both endogenous appropriation strategies and the strategic use of patents. I assume that each innovation along a standard quality-ladder is comprised of many individual components. Successful innovators choose a proportion of these components to patent and the corresponding proportion that remain secret. Both protection methods are imperfect, and innovators face a trade-off between protection against imitation by current quality followers and protection against subsequent quality innovations. Similar to Kultti et al. (2007) and Kwon (2012), patenting requires disclosure of underlying technical information in exchange for a probabilistic right to exclude imitators from using patented components. Following Dinopoulos and Syropoulos (2007), innovators may also invest resources in RPAs in order to raise the cost of competitors' innovation efforts. However, I explicitly connect the efficacy of RPAs investment to an innovator's endogenous patenting decision. Specifically, I assume that a firm's ability to impede competing innovation through certain types of RPAs, namely patent infringement litigation, depends upon its chosen patent portfolio. Secrecy, on the other hand, provides no formal protection against subsequent innovation. However, secrecy allows firms to avoid information disclosure and prevents imitation as long as this information does not “leak” to competitors.

The model provides a scale-effects free setting featuring fully endogenous growth and tractable analytics to examine how endogenous appropriation decisions influence our evaluations of intellectual property rights policy. For example, I show that strengthening patent protection, modeled as an increase in the probability of enforcement, generates two competing effects on the economy's rate of innovation. First, stronger patents produce the standard “reward effect” of increasing innovator profits and R&D incentives. However, as in Suzuki (2015), this effect is dampened since innovators rely only partially on patents in their overall appropriation strategy. Second, when patent protection is strengthened, firms endogenously increase the proportion of innovation components under patent to take advantage of the more secure monopoly rights they provide. This endogenous shift into patents increases the efficacy of incumbents' RPAs, effectively increasing the cost of R&D. For a realistic range of parameter values, I show that the net effect is a reduction in the innovation rate despite the overall increase in patenting, consistent with the observed patent puzzle.

In contrast, I find that strengthening the legal protection of trade secrets stimulates economic growth. Similar to case of strengthened patent protection, the enhanced security of trade secrets increases overall appropriability and stimulates R&D through a reward effect. Additionally, strengthening trade secret security generates competing effects on firm patenting be-

havior. On the one hand, making secrecy a more attractive appropriation mechanism creates a direct incentive to patent less. On the other hand, innovators are motivated to counteract the increase in competitors' R&D by increasing patent utilization to more effectively block innovation through RPAs. While the net effect on patenting behavior is ambiguous in general, I show that the overall effect of stronger trade secret protection is always growth enhancing. Although the policy change decreases the overall transfer of information to competitors since leakage rates are reduced, it generates the more typical policy trade-off of stimulating innovation by allowing for increased market power of innovators. Moreover, the model's predictions are consistent with the available empirical evidence that the strengthened legal protection of trade secrets in the U.S. has been associated with an increase in R&D expenditure and a decrease in patenting (Png, 2017a; 2017b).

Finally, I compare these intellectual property rights policy changes against the most common innovation enhancing policy examined in endogenous growth models, an R&D subsidy. While subsidizing R&D does not directly influence the relative security of patents and secrecy, it still generates important effects through endogenous firm appropriation strategies. Specifically, innovators respond to the increased threat of replacement via further innovation by increasing patent utilization to raise the cost of R&D. Although this shift into patenting only partially counteracts the growth effect of R&D subsidies and they can still be welfare improving, the resulting increase in the effectiveness of RPAs creates inefficiency. Indeed, numerical simulations of the model suggest that intellectual property rights policy can be used to generate an equivalent increase in the equilibrium growth rate, without the associated increase in resource wasting RPAs. In particular, simulations suggest that weakening patent protection generates the largest increase in social welfare since it stimulates innovation while limiting the negative effects of monopoly power and the use of RPAs.

The remainder of this paper is organized as follows: Section 2 presents the theoretical model. Section 3 examines the comparative statics of changes to R&D subsidies, patent strength, and trade secret security. I explore the welfare implications of the model in Section 4. Section 5 describes the calibration of the model and presents the results of numerical simulations. Section 6 concludes.

2. The model

2.1. Innovation and appropriation

The economy consists of a unit continuum of structurally identical industries indexed by $\omega \in [0, 1]$. In standard quality ladder fashion, innovation within each industry improves the quality of its existing product by a constant step size of $\lambda > 1$. In each industry ω and time t , there exists a single leading firm that has successfully innovated the current state of the art product with quality $q_l(\omega, t) = \lambda^{k(\omega, t)}$ by increasing quality over the previous $k(\omega, t) - 1$ iteration. Innovations are comprised of a unit measure of components. Following Suzuki (2015), each new leader determines an appropriation strategy as soon as innovation occurs by choosing a proportion $\mu(\omega, t) \in [0, 1]$ of these components to patent and the corresponding $1 - \mu(\omega, t)$ proportion to keep secret. I assume that patenting is costless.²

As is standard, I assume that competing, or challenger, firms within an industry have full access to the technology of the previous quality iteration, $k - 1$. In addition, challengers can increase their quality above the $k - 1$ product version if they are able to incorporate components of the latest innovation into their product. More specifically, letting $\theta_j(t) \in [0, 1]$ denote the proportion of components that challenger j in a typical industry has imitated at time t , the corresponding industry leader's actual quality advantage over challenger j is defined as

$$\eta_j(\theta, t) \equiv \frac{q_l(t)}{q_j(\theta, t)} = \frac{\lambda}{1 + (\lambda - 1)\theta_j(t)}. \quad (2.1)$$

Note that the leader enjoys the full λ quality lead when $\theta = 0$ and that the products have identical quality when the challenger has imitated all of the innovation with $\theta = 1$.

Patents grant an imperfect legal right to exclude challengers from imitating patented components, but require disclosure of the components' underlying technical information. There exists a probability $\delta \in (0, 1)$ that the leader's patents will not be effectively enforced. In this case, the set of patented components, $\mu(t)$, can be freely imitated by all challengers in the industry. I interpret δ as a broad measure of patent strength that distills all relevant aspects of the imperfect protection from imitation provided by patents. For example, δ can be viewed to reflect the overall threat of imitation due to the required disclosure of information coupled with limited patent breadth, length, and judicial enforcement of legal rights. Thus, $\delta = 0$ represents perfect patent protection and $\delta = 1$ represents nonexistent protection. Although secrecy does not require information disclosure, challengers are free to imitate the leader's $1 - \mu(t)$ set of secret components if they can discover how to incorporate the components into their product. I assume that all trade secrets face a constant probability $\ell \in (0, 1)$ of this information leakage to challengers.³ Directly analogous to δ , I interpret ℓ broadly as a measure of the

² Although clearly a simplification, available survey evidence suggests that the cost of patenting is typically not the determining factor in firms' patenting decision, especially for larger firms. For example, in a survey of over 1,000 U.S. manufacturing firms conducted by Cohen et al. (2000), only 36.7% of firms include application cost among their "reasons not to patent" and only 15.7% indicate it to be their most important reason not to patent.

³ I assume that all challengers have access to secret information in the event of a leak. However, as will be clear in the following sections, this is equivalent to interpreting ℓ as the aggregate probability that secret information leaks to a single challenger since price competition within industries implies that leader pricing and profit depend only on its quality lead over its nearest challenger.

imperfect security of trade secrets that encapsulates the overall threat of information leakage due to reverse engineering, espionage, and employee turnover.

Similar to Suzuki (2015), the expected proportion of an innovation that is imitated by challengers in a typical industry is determined by the strength of protection against imitation offered by patents (δ), secrecy (ℓ), and the leader's choice of appropriation strategy (μ). For any choice of $\mu(t)$, the expected proportion of imitated components is given by

$$\hat{\theta}(\mu, t) = \delta\mu(t) + \ell(1 - \mu(t)) = \ell + \mu(t)(\delta - \ell). \quad (2.2)$$

In Suzuki's model however, the incentive to patent stems solely from the assumption that patents provide greater protection from imitation than secrecy. Leaders choose their appropriation strategy to balance this incentive against an assumed positive and convex cost of patenting components, but they do not consider any dynamic effect that their patenting decision has on competing innovation. In contrast, I allow each leader to determine their appropriation strategy in order to optimize protection against the two threats to its position as industry leader: imitation and subsequent innovation by challenger firms. To reflect available empirical evidence, I assume throughout the analysis that $\delta > \ell$ so that secrecy offers superior protection from imitation. However, patents serve a blocking function: each leader leverages its patent portfolio to impede the innovative efforts of its challengers in order to prolong its market dominance.⁴

More specifically, challengers in each industry participate in R&D races to innovate the next quality improvement and supplant the current leader. Following Dinopoulos and Syropoulos (2007), market leaders invest resources in rent protection activities (RPAs) to raise the cost of this challenger innovation. For example, RPAs may include lobbying the government for favorable R&D policies, designing product attributes that make external improvements more difficult, and litigating patent infringement cases.⁵ The presence of RPAs creates an innovation contest between challengers and the leader within each industry. A challenger j that invests $R_j(t)$ in research at time t successfully innovates with instantaneous probability $I_j(t) = R_j(t)/D(t)$, where $D(t)$ captures the difficulty of R&D due to the leader's use of RPAs.

As in the literature, this difficulty depends on the endogenous intensity of investment in RPAs by the leader, denoted $X(t)$. However, I connect the efficacy of this investment to the leader's endogenous patenting decision. I define R&D difficulty at time t by,

$$D(t) \equiv \kappa X(t)[\alpha\mu(t) + (1 - \alpha)], \quad (2.3)$$

where $\kappa > 0$ and $0 < \alpha \leq 1$. The parameter α controls the strength of the blocking function of patents by determining the degree of complementarity between RPAs and patenting. For example, a firm's ability to effectively impede competitor innovation through investment in patent infringement litigation clearly depends on its patent portfolio. A more comprehensive set of patents may strengthen the legal argument that a rival innovation constitutes infringement or make it more costly for challengers to "invent around" existing patents to avoid potential infringement lawsuits. On the other hand, the effect of other types of RPAs, such as government lobbying efforts, may not be directly impacted by the use of patents or secrecy. The assumption that $0 < \alpha \leq 1$ ensures that patenting offers a net positive blocking function through its complementary effect on certain types of RPAs, but is flexible enough to allow for the marginal effect of RPAs to be partially independent of the leader's patent, secrecy choice.⁶

The industry wide innovation rate is obtained by summing across all challengers,

$$I(t) = \sum_j I_j(t) = \frac{R(t)}{\kappa X(t)[\alpha\mu(t) + (1 - \alpha)]}, \quad \text{where} \quad R(t) = \sum_j R_j(t). \quad (2.4)$$

As will be clarified below, both $X(t)$ and $R(t)$ are proportional to the size of the population in equilibrium, while $\mu(t)$ is constant. This implies that the model does not exhibit scale effects and the equilibrium rate of innovation is constant and independent of population size.

2.2. Households

The economy is populated by a unit continuum of identical households. Each household is a dynastic family of infinitely lived members that begins with a single member at $t = 0$ and grows at the common rate $n > 0$. The population of the economy at time t equals the size of each household, and is given by $N(t) = e^{nt}$. Each household maximizes the discounted utility

$$U = \int_0^\infty e^{-(\rho-n)t} \ln(u(t)) dt, \quad (2.5)$$

⁴ For example, of the over 1,000 U.S. manufacturing firms surveyed by Cohen et al. (2000), 51% view secrecy as an effective appropriation mechanism for product innovations compared to 34.8% for patents. For process innovations, 50.6% indicate secrecy is effective against just 23.3% for patents. When asked to indicate the reasons that they chose to apply for patents, 81.8% of firms indicate the blocking function of patents. See Arundel (2001), Cohen et al. (2000), Cohen et al. (2002), Hall et al. (2013) for further evidence.

⁵ For a detailed discussion of different types of RPAs and their importance, see Şener (2006), Şener (2008), Dinopoulos and Syropoulos (2007).

⁶ The case of $\alpha < 0$ corresponds to a negative impact of patenting on RPAs, which is equivalent to a positive relative blocking effect of using secrecy. Although certain types of RPAs may be more effective under secrecy, I assume an overall net positive blocking effect of patents throughout the analysis.

where $\rho > n$ is the subjective discount rate. Per capita sub-utility at time t is defined as

$$\ln(u(t)) = \int_0^1 \ln \left[\sum_k \lambda^k y(k, \omega, t) \right] d\omega, \quad (2.6)$$

where $y(k, \omega, t)$ denotes the quantity consumed of a product that has experienced k quality improvements in industry $\omega \in [0, 1]$ at time t .

Households maximize (2.5) by allocating per capita consumption expenditure $c(t)$ given prices at time t . Quality adjusted products within each industry are perfect substitutes, so households purchase only the product with the lowest quality adjusted price. Since products enter utility symmetrically, consumers optimally spread expenditure evenly across each industry. Simplifying notation, demand for the good with the lowest quality adjusted price in a typical industry is

$$y(t) = \frac{c(t)N(t)}{p(t)}, \quad (2.7)$$

where $p(t)$ is the market price of the good. Given (2.7), maximizing (2.5) subject to the standard intertemporal budget yields

$$\frac{\dot{c}(t)}{c(t)} = r(t) - \rho, \quad (2.8)$$

where $r(t)$ is the instantaneous market interest rate.

2.3. Labor and production

In each industry, labor is used for three separate tasks: R&D, RPAs, and the manufacture of final goods. Following [Dinopoulos and Syropoulos \(2007\)](#), I assume that labor is partitioned into two categories: general purpose labor that may be employed in either manufacturing or R&D and specialized labor that may only be used in RPAs.⁷ A constant proportion $s > 0$ of the population $N(t)$ are specialized, while the remaining $1 - s$ proportion are general purpose workers. Both factors grow at the overall population growth rate of n .

I normalize the wage rate of general purpose labor to unity and assume that one unit of labor produces one unit of the final good in each industry. In order to ensure that they offer the lowest quality adjusted price, the leader charges the optimal limit price equal to its quality lead over challengers, $p(\theta, t) = \eta(\theta, t)$. Given (2.7), the corresponding quantity sold is $y(\theta, t) = c(t)N(t)/\eta(\theta, t)$ and employment in production is $n_y(\theta, t) = c(t)N(t)(\lambda^{-1} + (1 - \lambda^{-1})\theta(t))$. Note that, for a given level of per capita consumption expenditure c , production employment increases in θ and approaches its competitive level of cN as θ goes to 1. The flow profits of the leader, gross of investment in RPAs, are

$$\pi(\theta, t) = p(\theta, t)y(\theta, t) - y(\theta, t) = c(t)N(t)(1 - \lambda^{-1})(1 - \theta(t)) = \pi_0(t)(1 - \theta(t)), \quad (2.9)$$

where $\pi_0(t)$ denotes the maximum possible flow profits at time t corresponding to a full λ lead with $\theta = 0$. That is, $\pi_0(t)$ represents the flow profits associated with perfect intellectual property protection as typically assumed in the literature. As the ability to exclude challengers from using an innovation decreases, θ goes to 1 and realized flow profits go to their competitive level of zero. Given (2.2), expected flow profits as a function of the leader's appropriation strategy are

$$\hat{\pi}(\mu, t) = \pi_0(t)[1 - \hat{\theta}(\mu, t)] = \pi_0(t)[1 - \ell - \mu(t)(\delta - \ell)]. \quad (2.10)$$

Exploiting symmetry across industries, total employment in production is

$$n_y(t) = c(t)N(t)[\lambda^{-1} + (1 - \lambda^{-1})\hat{\theta}(\mu, t)]. \quad (2.11)$$

Both R&D and RPAs are produced under constant returns to scale.⁸ Let $w_x(t) > 0$ denote the relative wage rate of specialized labor and β_x denote the unit-labor requirement in RPAs production. The corresponding cost function of RPAs is $w_x(t)\beta_x X(t)$. Similarly, letting β_R denote the unit-labor requirement in R&D production, the cost function associated with R&D is $\beta_R R(t)$. Total employment in R&D and RPAs respectively are given by

$$n_R(t) = \beta_R R(t), \quad n_x(t) = \beta_x X(t) \quad (2.12)$$

2.4. Stock market valuations

Let $V_c(t)$ denote the value of a challenger in a typical industry, $V(t)$ denote the value of successfully innovating to become the leader in this industry, and $0 \leq \sigma < 1$ denote the subsidy rate for R&D investment. The no-arbitrage condition associated with a challenger's R&D investment implies that a stock issued by a challenger must provide the same expected return as

⁷ Intuitively, we may interpret specialized workers as lawyers and lobbyists that are contracted from outside of the firm, while R&D and manufacturing use "in-house" labor. This labor partition creates a clear connection between consumption and RPAs in equilibrium, and allows for direct comparison with existing treatments. In [Appendix A](#), I show that the model's central results hold in a setting where labor is fully mobile across all activities.

⁸ This assumption is made for simplicity and follows [Dinopoulos and Syropoulos \(2007\)](#) and [Davis and Šener \(2012\)](#). The model's principle results continue to hold under decreasing returns to scale in R&D and/or RPAs.

a fully diversified investment of equal size. Over an interval of time dt , a $V_c(t)$ size investment in challenger j generates a realized gain of $V(t)$ with the probability of successful innovation $I_j(t)dt$. However, with probability $(1 - I_j(t)dt)$ innovation does not occur and there is an associated capital loss of $dV_c(t) = \dot{V}_c(t)dt$. Regardless of the outcome of its R&D effort, the firm incurs R&D cost equal to $(1 - \sigma)\beta_R R_j(t)dt$.

The overall equity return of these components must equal the risk-free return of $r(t)V_c(t)dt$. Thus, the corresponding no-arbitrage condition is

$$r(t)V_c(t)dt = I_j(t)V(t)dt - (1 - \sigma)\beta_R R_j(t)dt + (1 - I_j(t)dt)\dot{V}_c(t)dt. \quad (2.13)$$

Free-entry into R&D implies that $\dot{V}_c = V_c = 0$. Taking limits at dt goes to zero implies the following free-entry condition, $I_j(t)V(t) = (1 - \sigma)\beta_R R_j(t)$, which equates the expected return of R&D efforts to their cost. Using the definition of $I_j(t)$, the free-entry condition can be rewritten as

$$V(t) = (1 - \sigma)\beta_R K X(t)[\alpha\mu(t) + (1 - \alpha)], \quad (2.14)$$

which relates the value of an innovation to the cost of conducting R&D implied by the leader's patenting and RPAs decisions.

Similarly, the expected return from holding $V(t)$ of stock in an industry leader over interval dt is the expected profit flow $\hat{\pi}(t)dt$, minus the cost of RPAs $w_x(t)\beta_x X(t)dt$ and the $I(t)dt$ probability of capital loss $V(t)$ from replacement, plus the change in valuation $\dot{V}(t)dt$ if the firm remains the leader. The corresponding no-arbitrage condition is

$$rV(t)dt = \hat{\pi}(t)dt - w_x(t)\beta_x X(t)dt - I(t)V(t)dt + (1 - I(t)dt)\dot{V}(t)dt. \quad (2.15)$$

2.5. Equilibrium

I now solve the model for a steady state equilibrium in which $\mu(t)$, $I(t)$, $w_x(t)$, and $c(t)$ are constant, $X(t)$, $\pi_0(t)$, and $V(t)$ grow at the rate of population growth n , labor markets clear, the free-entry condition of (2.14) holds, and leaders choose $X(t)$ and $\mu(t)$ to maximize their value given by (2.15). Henceforth, I drop the time index for all variables that are constant in equilibrium. In accordance with available empirical evidence, I focus attention in the main text to the case where neither appropriation mechanism is dominant, and innovations are protected with a mix of patents and secrecy.⁹ I establish the equilibrium by deriving two equilibrium conditions in μ and I .

2.5.1. Equilibrium condition 1: the patenting condition

To understand the set of parameters that is consistent with $0 < \mu < 1$ in equilibrium, first recall that both $X(t)$ and μ involve an upfront cost in exchange for extending the expected duration of a firm's leader position by reducing the probability of replacement through subsequent innovation. In the case of RPAs, there is a direct marginal cost of $w_x\beta_x$. From (2.10), patenting also carries a constant marginal cost at time t in the form of a reduction in expected profit flow of $\pi_0(t)(\delta - \ell)$ due to the inferior protection from imitation provided by patents. The benefit of a marginal increase in $X(t)$ and μ respectively are $-\frac{\partial I}{\partial X(t)}V(t)$ and $-\frac{\partial I}{\partial \mu}V(t)$. Given the specification of I in (2.4),

$$-\frac{\partial I}{\partial X(t)} = \frac{I}{X(t)}, \quad -\frac{\partial I}{\partial \mu} = \frac{\alpha I}{[\alpha\mu + (1 - \alpha)]}. \quad (2.16)$$

Since $-\frac{\partial I}{\partial X(t)} \rightarrow \infty$ as $X(t) \rightarrow 0$ for any $I > 0$, it is clear that $X(t) > 0$ is always optimal. We have the following optimality condition equating the marginal cost and benefit of RPAs,

$$IV(t) = w_x\beta_x X(t). \quad (2.17)$$

This implies that the expected loss due to replacement equals total expenditure on RPAs at the optimal level of $X(t)$.

Similarly, for any interior patenting solution, $0 < \mu < 1$, we have the following first order necessary condition

$$\frac{\alpha IV(t)}{[\alpha\mu + (1 - \alpha)]} = \pi_0(t)(\delta - \ell). \quad (2.18)$$

To proceed, take limits as $dt \rightarrow 0$ in (2.15), use (2.17), and note that $\dot{V}(t)/V(t) = n$ and $r = \rho$ in equilibrium to rewrite the leader's no-arbitrage condition as

$$V(t) = \frac{\pi_0(t)[1 - \hat{\theta}(\mu)]}{\rho - n + 2I}. \quad (2.19)$$

In equilibrium, leaders discount the flow profits implied by a full λ quality lead due to the loss in profits from imperfect appropriation, the cost of RPAs according to (2.17), and the constant probability of replacement. Using (2.19) allows us to rewrite the necessary condition associated with $0 < \mu < 1$ as

$$I = \frac{(\rho - n)(\delta - \ell)[\alpha\mu + (1 - \alpha)]}{\alpha(1 - \ell) - 2(\delta - \ell)(1 - \alpha) - 3\alpha\mu(\delta - \ell)}. \quad (2.20)$$

⁹ I examine the case where patents strictly dominate secrecy ($\mu = 1$) in Appendix A, and show that the model replicates Dinopoulos and Syropoulos (2007) in the special case of perfect patent protection, $\delta = 0$.

Since $(\delta - \ell) > 0$, the numerator of (2.20) is always positive. The following restriction is sufficient for $0 < \mu < 1$ to be consistent with $I > 0$ in equilibrium,

Assumption 1.

$$0 < \mu < \frac{\alpha(1 - \ell) - 2(\delta - \ell)(1 - \alpha)}{3\alpha(\delta - \ell)} < 1.$$

Under Assumption 1, Eq. (2.20) defines the equilibrium patenting condition that expresses the relationship between $I > 0$ and optimal patenting behavior. The patenting condition does not depend on consumption because both the benefit of patenting (reduced risk of replacement) and the cost of patenting (reduced profit flows) are proportional to c . Since leaders have a stronger incentive to use patents to raise the cost of R&D when they face a higher risk of replacement, the innovation rate specified by the patenting condition is strictly increasing in μ . Assumption 1 ensures that the parameters that control the relative cost and benefit of patenting imply that leaders are incentivized to choose a mix of patenting and secrecy in equilibrium. For example, for any $I > 0$, leaders have the greatest incentive to patent when $\alpha = 1$ and the blocking effect of patents is strongest. In this case, Assumption 1 requires a sufficiently large cost of patenting $(\delta - \ell) > (1 - \ell)/3$ to ensure that leaders always choose $\mu < 1$. On the other hand, as α approaches 0, the blocking effect disappears and the cost of patenting $(\delta - \ell)$ must also go to zero to ensure that leaders remain incentivized to choose $\mu > 0$.

Observe that the patenting condition implicitly defines a lower bound for the innovation rate to be consistent with $\mu > 0$. That is, as $\mu \rightarrow 0$, (2.20) implies that $I \rightarrow I_{\min}$, where

$$I_{\min} \equiv \frac{(\rho - n)(\delta - \ell)(1 - \alpha)}{\alpha(1 - \ell) - 2(\delta - \ell)(1 - \alpha)} \geq 0. \quad (2.21)$$

Intuitively, $I_{\min}$ defines the lowest rate of replacement for which leader's are incentivized to choose $\mu > 0$, for a given set of parameters. Note that $I_{\min}$ is strictly decreasing in α and $I_{\min} \rightarrow 0$ as $\alpha \rightarrow 1$. In other words, a stronger blocking effect due to greater complementarity between patenting and RPAs implies that leaders are incentivized to begin patenting at lower rates of innovation. Similarly, Assumption 1 directly defines an upper bound for patenting in order for the innovation rate implied by the patenting condition to be finite. Note that this upper bound is not a restriction on leaders' ability to choose μ ; it is a consequence of their optimizing behavior given the parameter restriction in Assumption 1. That is, for a given set of parameters, leaders have the strongest incentive to patent when $I \rightarrow \infty$, and (2.20) implies that the optimal choice of μ approaches $[\alpha(1 - \ell) - 2(\delta - \ell)(1 - \alpha)]/3\alpha(\delta - \ell) \equiv \mu_{\max}$. For future use, note that $\mu_{\max} < 1$ under Assumption 1 and that $I \rightarrow \infty$ as $\mu \rightarrow \mu_{\max}$.

2.5.2. Equilibrium condition 2: the resource constraint

General purpose labor market clearing requires that the supply, $(1 - s)N(t)$, must equal the demand for manufacturing and R&D labor given by (2.11) and (2.12). Similarly, specialized labor market clearing requires that $sN(t) = \beta_x X(t)$. Using the definition of I from (2.4), these two labor market clearing conditions can be combined into a single equilibrium resource constraint as a function of μ , I , and c ,

$$(1 - s) = \frac{c}{\lambda} [1 + (\lambda - 1)(\ell + \mu(\delta - \ell))] + \frac{\beta_{RK}[\alpha\mu + (1 - \alpha)]s}{\beta_x} I. \quad (2.22)$$

Eq. (2.22) illustrates that the standard resource allocation trade-off between consumption and innovation is preserved with the inclusion of endogenous patenting. Indeed, an increase in patent utilization "tightens" the resource constraint since it raises the effective labor requirement for innovation and increases the manufacturing labor needed to maintain a constant level of per capita consumption expenditure.

The ratio of the free-entry condition (2.14) and the optimality condition for RPAs (2.17) provides an expression for the equilibrium innovation rate in terms of the effective relative cost of RPAs and R&D,

$$I = \frac{w_x \beta_x}{(1 - \sigma) \beta_{RK} [\alpha\mu + (1 - \alpha)]} \quad (2.23)$$

The relative wage of specialized labor adjusts so that the supply and demand of specialized labor equalize. Using the optimality condition for RPAs (2.17) and the labor market clearing condition for specialized workers, we have $w_x = IV(t)/sN(t)$. Substituting w_x into (2.23) yields the equilibrium relative profitability condition in terms of μ , I , and c ,

$$c = \frac{(1 - \sigma) \beta_{RK} [\alpha\mu + (1 - \alpha)] s \lambda (\rho - n + 2I)}{\beta_x (\lambda - 1) [1 - \hat{\theta}(\mu)]}. \quad (2.24)$$

Finally, combine (2.22) and (2.24) to eliminate c ,

$$I = \frac{\beta_x (1 - s) (\lambda - 1) (1 - \hat{\theta}(\mu)) - (\rho - n) (1 - \sigma) \beta_{RK} [\alpha\mu + (1 - \alpha)] s (1 + (\lambda - 1) \hat{\theta}(\mu))}{\beta_{RK} [\alpha\mu + (1 - \alpha)] s [2(1 - \sigma) (1 + (\lambda - 1) (\ell + \mu(\delta - \ell))) + (\lambda - 1) (1 - \hat{\theta}(\mu))]} \quad (2.25)$$

Condition (2.25) defines the equilibrium resource constraint after incorporating the allocative effect of the relative profitability of R&D and RPAs. Since the blocking function of patenting increases the effective resource cost of innovation according

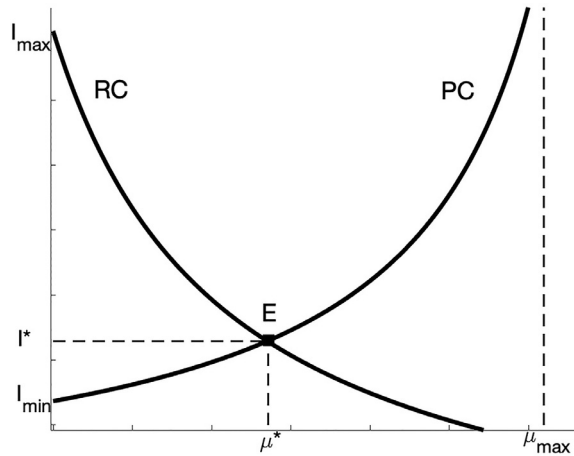


Fig. 1. Equilibrium.

to (2.3), Eq. (2.25) specifies a downward sloping relationship between I and μ . In addition, note that (2.25) provides an upper bound on the rate of innovation the economy's resources can support, given the presence of RPAs. That is, as $\mu \rightarrow 0$, (2.25) implies that $I \rightarrow I_{\max}$, where

$$I_{\max} \equiv \frac{(1-s)\beta_x(\lambda-1)(1-\ell) - (1-\sigma)\beta_{RK}(1-\alpha)s(\rho-n)(1+(\lambda-1)\ell)}{s\beta_{RK}(1-\alpha)[(\lambda-1)(1-\ell) + 2(1-\sigma)(1+(\lambda-1)\ell)]}. \quad (2.26)$$

Observe that $I_{\max}$ is strictly increasing in α , with $I_{\max} \rightarrow \infty$ as $\alpha \rightarrow 1$. Intuitively, greater complementarity between patents and RPAs implies that RPAs are less effective in the absence of patents, and that the economy can support a higher innovation rate as $\mu \rightarrow 0$.

2.5.3. The steady state equilibrium

The model's equilibrium is determined by solving the patenting condition (PC) given by (2.20) and the resource constraint (RC) given by (2.25) for μ and I . Fig. 1 illustrates the equilibrium by graphing the PC and RC in (μ, I) space. As depicted in the figure, the following additional parameter restriction is required to guarantee an intersection of the PC and RC curves,

Assumption 2.

$I_{\max} > I_{\min} \geq 0$, as defined in (2.26) and (2.21) respectively.

Note that Assumption 2 is always satisfied in the special case of $\alpha = 1$, since $I_{\max} \rightarrow \infty$ and $I_{\min} \rightarrow 0$ as $\alpha \rightarrow 1$.

Under Assumptions 1 and 2, a unique steady state equilibrium exists in which $I > 0$ and $0 < \mu < 1$.¹⁰ After establishing I and μ , all other endogenous variables can be determined. I follow standard practice in quality ladder models and define economic growth as the rate of growth of per capita sub-utility $u(t)$. As I show in Section 4, the present model inherits the usual feature that the equilibrium rate of economic growth is proportional to the rate of innovation,

$$g \equiv \frac{\dot{u}(t)}{u(t)} = \ln(\lambda)I, \quad (2.27)$$

where $\ln(\lambda) > 0$ represents the utility benefit of each innovation. Furthermore, since new innovations arrive at the constant rate $I > 0$ in equilibrium and each leader patents a constant proportion $0 < \mu < 1$ of the innovation's components, the equilibrium exhibits a constant arrival rate of new patents equal to

$$\psi \equiv \mu I. \quad (2.28)$$

To avoid confusion, I refer to ψ as the "patent rate" and refer to each firm's choice of patenting in its appropriation strategy, μ , as firm "patent utilization."

¹⁰ As drawn, the RC curve intersects the horizontal axis at some $\mu < \mu_{\max} < 1$. While one can confirm from (2.25) that this represents one possible case, note that this is not required for an equilibrium to exist. Even if parameters are such that the horizontal intercept is to the right of $\mu = 1$, the optimal patenting behavior embedded in the PC curve ensures that $I \rightarrow \infty$ as $\mu \rightarrow \mu_{\max}$ and that the two curves always intersect at $0 < \mu^* < \mu_{\max}$.

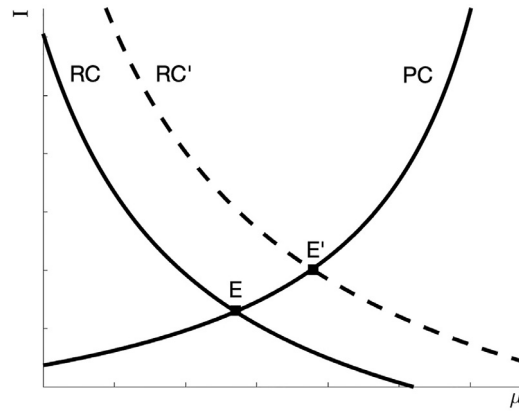


Fig. 2. Subsidy to R&D.

3. Policy implications

In this section I examine how policy changes influence the model's steady state equilibrium. As in [Dinopoulos and Syropoulos \(2007\)](#), the model does not exhibit transitional dynamics, and immediately shifts to its new steady state.¹¹ I analyze comparative statics exercises with respect to changes in three distinct policy parameters: the R&D subsidy rate σ , patent strength δ , and trade secret security ℓ . My primary focus is to understand how each of these policies influence the general equilibrium relationship between endogenous appropriation strategies and economic growth.

3.1. Subsidy to R&D

The direct effect of subsidizing R&D is an increase in the relative profitability of R&D. This directly implies an increase in the resources allocated to R&D corresponding to a rightward shift in the resource constraint in [Fig. 2](#).

Although the R&D subsidy does not directly impact the patenting condition, the movement along this curve results in a greater level of firm patent utilization in the new equilibrium. Intuitively, leaders respond to the increased threat of replacement by bolstering RPAs with a larger patent portfolio. This endogenous response partially offsets the positive innovation effect of the subsidy. This finding is summarized in the following proposition,

Proposition 1. *Subsidizing R&D expenditure increases firm patent utilization, increases the innovation rate, and increases the patent rate. That is, $\frac{\partial \mu}{\partial \sigma} > 0$, $\frac{\partial I}{\partial \sigma} > 0$, and $\frac{\partial \psi}{\partial \sigma} > 0$.*

3.2. Patent strength

Strengthening patent protection (decreasing δ) introduces two main effects. First, the increase in the overall ability to protect profit flows from imitation increases the reward from innovating successfully. This incentivizes challengers to increase R&D investment, and reflects the traditional motivation for a strong patent regime. However, as emphasized by [Suzuki \(2015\)](#), the strength of this reward effect depends upon the importance of patents in firms' appropriation strategies. This is represented in [Fig. 3](#) by a nonparallel rightward shift in the resource constraint, where the size of the shift is greater for larger μ . Second, strengthening patent protection reduces the effective cost of patenting relative to secrecy. This implies that leaders increase the proportion of patented components for any rate of innovation, and is represented by a rightward shift in the patenting condition.

As illustrated in [Fig. 3](#), both shifts point to an increase in patent utilization. The change in the innovation rate is determined by the relative importance of the reward effect and the blocking effect of the shift into patenting. In traditional treatments where patents are the sole appropriation mechanism considered, the full reward effect is always present and strengthening patent protection increases the equilibrium innovation rate. In Suzuki's treatment where firms choose a mix of secrecy and patents, the size of the reward effect depends on firms' patent utilization. Since Suzuki does not allow firms to consider the blocking effect of patenting, he finds that strengthening patent protection stimulates innovation in cases where relatively weak trade secret protection causes firms to rely primarily on patents.

In the present model however, the incentive to patent is directly linked to their usefulness as an innovation deterrent. Since firms patent specifically to take advantage of their blocking function, the size of the reward effect is necessarily connected to the size of the blocking effect. Consequently, one can show that the blocking effect *always* dominates the reward

¹¹ Intuitively, this is because the model's key endogenous variables are all choice variables that instantaneously adjust to new steady state levels. See the Appendix of [Dinopoulos and Syropoulos \(2007\)](#) for details.

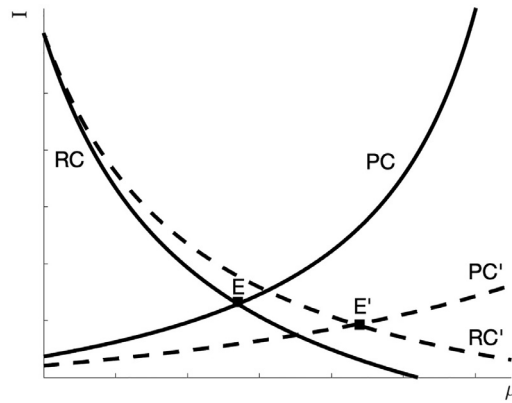


Fig. 3. Strengthening patent protection.

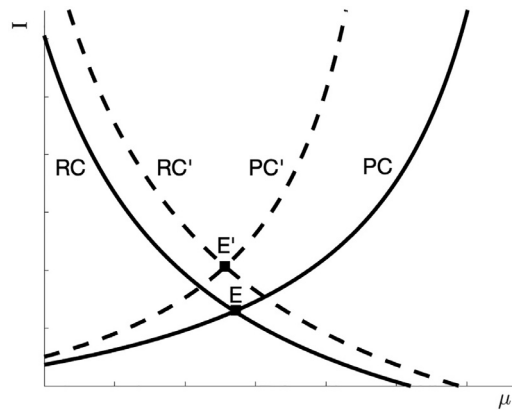


Fig. 4. Strengthening trade secret protection.

effect, and the innovation rate decreases as a result of strengthened patent protection. Indeed, this core result holds for any $0 < \alpha \leq 1$, as long as the model's parameters satisfy [Assumptions 1](#) and [2](#). That is, even in cases where the blocking effect of patenting is assumed to be weak, the corresponding optimal utilization of patents in firms' appropriation strategies implies a similarly weak reward effect.

Finally, note that the change to the equilibrium patent rate, ψ , resulting from strengthened patent protection depends on the relative magnitude of the decrease in the innovation rate and the increase in patent utilization in firm appropriation strategies. In general, the direction of this change is ambiguous, and I provide an example of each case in the numerical analysis of [Section 5](#). However, since μ always increases when patent protection is strengthened, transition to the new steady state equilibrium always requires that all existing industry leaders undertake a one-time patent increase. Although this occurs instantaneously in the context of the model, it represents an initial surge in new patents even in cases where the long-run equilibrium patent rate is ultimately lower. These findings are summarized in the following proposition,

Proposition 2 (The patent puzzle). *Strengthening the protection against imitation provided by patents increases patent utilization, decreases the innovation rate, and has an ambiguous effect on the patent rate. That is $\frac{\partial \mu}{\partial \delta} < 0$, $\frac{\partial I}{\partial \delta} > 0$, and $\frac{\partial \psi}{\partial \delta} > < 0$.*

Proof. See Appendix. $\square$

3.3. Trade secret protection

When secrecy is made a more effective appropriation mechanism, the cost of patenting and forgoing secrecy's superior protection from imitation increases. This incentives leaders to decrease patent utilization and rely more heavily on secrecy at any rate of innovation. Correspondingly, the patenting condition shifts to the left. Moreover, as in the case of strengthened patents, the overall increase in appropriability generates a reward effect which shifts the resource constraint to the right. Once again, the size of this reward effect depends upon the importance of secrecy in leaders' appropriation strategy. That is, it is larger for smaller μ .

As can be seen in [Fig. 4](#), these two shifts unambiguously increase the rate of innovation. However, the shifts generate opposing forces on the equilibrium level of patent utilization since movement along the optimal patenting condition reflects

the incentive to increase patenting in response to a greater threat of replacement. Indeed, I show in the Appendix that the change to μ is ambiguous in general, and depends on the relative security of secrecy and patents. Similarly, the change to the equilibrium patent rate is ambiguous, as it depends on the direction and magnitude of change in μ . We have the following,

Proposition 3. *Strengthening trade secret protection has an ambiguous effect on patent utilization, increases the innovation rate, and has an ambiguous effect on the patent rate. That is, $\frac{\partial \mu}{\partial \ell} > 0$, $\frac{\partial I}{\partial \ell} < 0$, and $\frac{\partial \psi}{\partial \ell} > 0$.*

Proof. See Appendix. $\square$

4. Welfare

As is standard in endogenous growth models, instantaneous per capita utility depends on three factors: product quality, per capita expenditure, and prices. In the present model however, prices depend on the strength of intellectual property protection and firms' endogenous appropriation decision since limit pricing implies $p = \eta$. From (2.6) and (2.7), instantaneous per capita utility can be written as $\int_0^1 \ln(\lambda^k c / \eta(\omega)) d\omega$. Exploiting symmetry across industries, there are four relevant categories of prices for a given equilibrium μ . 1) A proportion $\delta \ell$ of industries have experienced both trade secret leakage and unenforced patents. These products are priced competitively at $p = 1$ independent of μ . 2) $(1 - \delta)\ell$ products have experienced leakage but its patents have been enforced. These products are priced at $p = \lambda / (1 + (\lambda - 1)(1 - \mu))$. 3) $\delta(1 - \ell)$ products have not experienced leakage but its patents have not been enforced. The corresponding price is $p = \lambda / (1 + (\lambda - 1)\mu)$. 4) $(1 - \delta)(1 - \ell)$ of industries have experienced no imitation by challengers, and are priced at the full monopoly mark-up of $p = \lambda$. As shown in the Appendix, performing the required integration yields the following expression for per capita utility at time t ,

$$\ln(u(t)) = \ln(\lambda)I \cdot t + m(c, \mu), \quad \text{where} \quad (4.1)$$

$$m(c, \mu) \equiv \ln(c/\lambda) + \left[\delta \ell \ln(\lambda) + (1 - \delta)\ell \ln(1 + (\lambda - 1)(1 - \mu)) + \delta(1 - \ell) \ln(1 + (\lambda - 1)\mu) \right]. \quad (4.2)$$

The first term in (4.1) represents the welfare effect of growth in product quality due to innovation over time. Differentiating (4.1) with respect to time yields $g = \ln(\lambda)I$ as displayed in (2.27). Using (2.5), welfare discounted to time zero can be written as

$$(\rho - n)U = \frac{\ln(\lambda)}{\rho - n}I + m(c, \mu). \quad (4.3)$$

The $m(c, \mu)$ term represents welfare from consumption expenditure, taking into account the price effect of imperfect intellectual property rights. That is, the first term in $m(c, \mu)$ is the standard welfare effect of consumption expenditure given monopoly pricing. The bracketed term is unique to the present model, and represents the positive welfare effect of price reductions induced by imitation. Note that if intellectual property protection were perfect ($\delta = 0$, $\ell = 0$), $m(c, \mu) = \ln(c/\lambda)$, since monopoly pricing prevails for any μ . On the other hand, if intellectual property received no protection ($\delta = 1$, $\ell = 1$), $m(c, \mu) = \ln(c)$, since all products would be priced competitively at $p = 1$.

Although it is clear that $m(c, \mu)$ is strictly increasing in c , $m(c, \mu)$ can be non-monotonic in μ . Since $\delta > \ell$ by assumption, expected imitation $\hat{\theta}(\mu)$ is strictly increasing in μ . Correspondingly, expected prices always decrease in μ reflecting the positive disclosure effect of patenting. While this suggests that $m(c, \mu)$ is typically increasing in μ , the benefit to consumption smoothing across industries generated from log utility implies that $m(c, \mu)$ can begin to decrease in μ when μ is sufficiently high under certain parameter values. That said, most cases, including the parameters of the numerical analysis below, exhibit strictly increasing $m(c, \mu)$, and I interpret this to be the usual case. See the Appendix for further details.

I consider a social planner who chooses the levels of I and c to maximize social welfare in (4.3), subject to the resource constraint (2.22). As in Dinopoulos and Syropoulos (2007), this implicitly assumes that the social planner cannot tax (or otherwise control) the allocation of resources to RPAs, nor impact the pricing decisions of industry leaders. In the same spirit, I assume that the planner cannot directly control leaders' patenting decisions, which implies that patent utilization is determined by the optimality condition given by (2.20). The associated Lagrangian is

$$\mathcal{L}(c, I, \Omega) = \frac{\ln(\lambda)}{\rho - n}I + m(c, \mu) + \Omega \left[1 - s - \frac{c}{\lambda} [1 + (\lambda - 1)\hat{\theta}(\mu)] - \frac{\beta_R \kappa [\alpha \mu + (1 - \alpha)s] s}{\beta_x} \right], \quad (4.4)$$

where Ω is the Lagrange multiplier. Observe that the social planner's constraint reflects the traditional trade-off between the resources allocated to the production of final goods and R&D. Optimization results in the following expression that equates the social cost and return of R&D,

$$\begin{aligned} \frac{\beta_R \kappa [\alpha \mu + (1 - \alpha)s] s}{\beta_x} &= \frac{c}{\lambda} \frac{\ln(\lambda)}{\rho - n} [1 + (\lambda - 1)\hat{\theta}(\mu)] \\ &+ \frac{\partial \mu}{\partial I} \left[\frac{c}{\lambda} [1 + (\lambda - 1)\hat{\theta}(\mu)] \frac{\partial m}{\partial \mu} - \left(\frac{c}{\lambda} (\lambda - 1)(\delta - \ell) + \frac{\beta_R \kappa s \alpha I}{\beta_x} \right) \right]. \end{aligned} \quad (4.5)$$

Combining (2.14) and (2.19) yields the analogous expression for the market cost and return of R&D,

$$(1 - \sigma) \frac{\beta_R \kappa [\alpha \mu + (1 - \alpha)s]}{\beta_x} = \frac{c}{\lambda} \frac{\lambda - 1}{\rho - n + 2I} [1 - \hat{\theta}(\mu)]. \quad (4.6)$$

Comparing these expressions, it is clear that there are several reasons for the socially optimal and market levels of R&D to differ. First, the model inherits two effects directly from [Dinopoulos and Syropoulos \(2007\)](#) due to the models' shared expression for the contribution of economic growth to the present value of welfare, $\ln(\lambda)I/(\rho - n)$. The *monopoly distortion effect* is present because the social planner considers the utility benefit of an innovation $\ln(\lambda)$, while the market is concerned with the potential monopoly mark-up rate of $\lambda - 1$. Since $\lambda - 1 > \ln(\lambda)$, this effect implies that the market rate of R&D is too high. Next, the *intertemporal spillover effect* arises because the social planner discounts the benefits of each innovation by $\rho - n$, while the effective market discount rate of $\rho - n + 2I$ incorporates the expected loss due to replacement by rival innovation, as well the cost of deterring innovation through RPAs investment. This effect suggests that the market undertakes too little R&D.

In addition, the model generates three novel welfare effects. First, private researchers discount the stream of monopoly profits by $[1 - \hat{\theta}(\mu)]$ based on their imperfect ability to exclude competitors from imitating their innovation. On the other hand, the social planner scales the return to innovation by $[1 + (\lambda - 1)\hat{\theta}(\mu)]$, reflecting the beneficial impact of imitation on prices. This *imperfect appropriation effect* suggests that the private market undertakes too little R&D. Next, the social planner internalizes the feedback effect of the innovation rate on the patenting behavior of private firms. That is, the planner realizes that a consequence of more R&D is higher patent utilization, since $\partial \mu / \partial I > 0$ as firms choose patent utilization to maximize profits according to (2.20).

This generates two effects that are specific to the model's treatment of patents. The first term in the brackets of (4.5) represents the welfare effect of the change to imitation induced by increased patent utilization. As mentioned above, in most circumstances $\partial m / \partial \mu > 0$, and the increased imitation benefits consumers. I refer to this channel as the *disclosure effect*.¹² Finally, the last term in brackets represents the negative effect of the increased resource cost of R&D and consumption expenditure as a result of higher patent utilization. This effect unambiguously reduces the returns to innovation, and is dubbed the *resource wasting effect*. In contrast, private researchers completely ignore the impact of their R&D efforts on patenting behavior.

As is common in the endogenous growth literature, the market equilibrium can in principle deliver either too much or too little R&D from a social perspective. Unfortunately, the model is too complex to provide an explicit characterization of the circumstances that lead to each case. In general however, the combined impact of incorporating RPAs and imperfect intellectual property protection suggest that the model will tend towards an equilibrium with a suboptimally low rate of innovation. As shown in [Appendix A](#), the model replicates [Dinopoulos and Syropoulos \(2007\)](#) in the special case of perfect patent protection. Even in this case, [Dinopoulos and Syropoulos](#) show that the presence of RPAs alone implies that the market equilibrium rate of innovation is below the socially optimal level over a broad range of plausible parameters. Allowing weaker overall intellectual property protection amplifies this discrepancy through the imperfect appropriation effect. Indeed, it is clear that as $\delta, \ell \rightarrow 1$, market based R&D incentives collapse entirely.

This implies that improving social welfare will tend to require policy changes that increase the equilibrium rate of innovation. For example, an R&D subsidy can be implemented to equate the market and socially optimal level of R&D for given values of δ and ℓ . Moreover, the model offers two additional intellectual property rights policy levers to influence both the market and social returns to R&D. Each of these three policy options influences welfare through their primary impact on economic growth in addition to their more nuanced impact through changes to endogenous firm appropriation strategies. Importantly, this implies that these policies may have different welfare implications even when they share a common economic growth effect. In the following section, I examine the policy implications of the model through numerical analysis.

5. Numerical analysis

In all simulations, I use the following baseline parameters: $\rho = 0.07$, $n = 0.01$, $\beta_x = \kappa = 1$, and $s = 0.025$. As is common, I set ρ to 0.07 to reflect a 7% long run real return of the U.S. stock market, and $n = 0.01$ to approximate the average growth rate of the U.S. labor force over the past thirty years ([Bank, 2019](#)). Next, since β_x and κ impact the equilibrium of the model only through their value relative to β_R , I normalize them both to 1. I set $s = 0.025$ to generate a reasonable relative wage for specialized workers of about 1.5 as targeted in [Şener \(2008\)](#).

I set the parameters, λ , β_R , α , δ , and ℓ , in order to attain realistic values of price mark-ups over marginal cost, the proportion of patented components, and economic growth. First, in models with perfect patent protection, the size of each quality innovation λ corresponds directly to each leader's gross markup over marginal cost. Thus, λ is set to match evidence of mark-ups in the range of empirical estimates of 1.05 - 1.4. The most common choice in the endogenous growth literature seems to be a 25% mark-up, corresponding to $\lambda = 1.25$ ([Segerstrom, 2007](#); [Şener, 2008](#); [Davis and Şener, 2012](#)). In the present model, each leader's mark-up is determined by its effective quality lead over imitators, η , which depends on λ as well as δ , ℓ , and endogenous patent utilization μ . Following the literature, I target $\eta = 1.25$ corresponding to a 25%

¹² This social benefit of patenting captured by the model is closely related to the "contract theory" of patents, which emphasizes the positive effect of the disclosure function of the patent system. See [Denicolò and Franzoni \(2003\)](#) for a thorough discussion.

Table 1
Calibrated parameters.

	Strong ($\alpha = 0.67$)		Weak ($\alpha = 0.33$)	
	$\mu = 0.45$	$\mu = 1.0$	$\mu = 0.45$	$\mu = 1.0$
δ	0.408	0.365	0.344	0.335
β_R	170.3	108.7	128.7	106.9
λ	1.443	1.460	1.416	1.431

markup. Next, I follow [Segerstrom \(2007\)](#) and [Şener \(2008\)](#) and target a growth rate $g = \ln(\lambda)I$ of 0.5%, which is in line estimates of long-run growth attributable to technological advancements. Finally, as mentioned in the introduction, estimates for patent propensity, defined as the proportion of innovations for which a patent application is made, range from 30 to 55% ([Cohen et al., 2002](#); [Hall et al., 2014](#)). This is not a perfect analogue to the model's μ , but it helps to inform a reasonable target. I choose an intermediate target value of $\mu = 0.45$. In addition, for comparison with existing treatments that assume innovations are protected solely by patents, I examine a “patent dominant” case where $\mu = 1$ is optimal.¹³

Since firm patent decisions depend on the blocking effect of patents and the relative protection of patents and secrecy, α , δ , and ℓ are not separately identified without additional restrictions. To account for this, I fix ℓ and α to plausible values and treat δ as a free calibration parameter. While there is not an obvious empirical counterpart to the model's stylized representation of trade secret strength, [Cohen et al. \(2002\)](#) reports that unpatented product and process innovations in the U.S. enjoy an average duration of 2.80 and 3.37 years before imitation respectively. In order to be broadly consistent with this finding, I fix $\ell = 0.3$ throughout the main text. Similarly, it is difficult to identify a direct empirical analogue of the model's representation of the degree of complementarity between patenting and RPAs. On the other hand, patent litigation clearly comprises a major component of RPAs, and empirical studies indicate that patent litigation costs often amount to a substantial portion of total R&D spending for new innovators ([Lerner, 1995](#); [Hall et al., 2004](#); [Bessen and Meurer, 2013](#)). In order to explore how numerical results depend upon the strength of the blocking effect of patents, I examine two separate cases: a relatively “strong” blocking effect case with $\alpha = 0.67$ and a “weak” blocking effect case with $\alpha = 0.33$.

With all other parameters fixed, I jointly calibrate λ , β_R , and δ to match a growth rate of 0.5%, a 25% mark-up over cost, and the patent utilization target of either $\mu = 0.45$ or $\mu = 1$. The resulting parameters are summarized in [Table 1](#). First observe that, for each fixed value of α , moving from $\mu = 0.45$ to $\mu = 1$ requires a smaller calibrated value of δ because stronger protection from imitation is needed to incentivize the increase in patent utilization.¹⁴ Since the effective cost of R&D increases with patent utilization, β_R must decrease in order to maintain the targeted 0.5% initial growth rate. Notice that this difference is more pronounced when the blocking effect of patents is stronger. Similarly, λ increases in order to maintain the targeted 25% mark-up over cost given the increased utilization of patents, and their inferior protection from imitation relative to secrecy.

Next, consider how the calibrated parameters adjust moving from the strong to weak blocking effect case for each fixed patent utilization target. A weaker blocking effect requires a smaller δ so that relatively stronger imitation protection incentivizes the same level of patent utilization. This implies more overall protection from imitation, and λ must decrease to maintain the 25% mark-up. Finally, to maintain the targeted growth rate, β_R must decrease for two reasons. First, for any $\mu < 1$, note that (2.4) implies that a smaller α corresponds to a greater overall cost of R&D, ceteris paribus. Of course, the contribution of patents to the cost of R&D still decreases with α .¹⁵ Second, since the economy's growth rate is $g = \ln(\lambda)I$, the decrease in λ implies that a larger innovation rate is required to generate the same rate of growth.

5.1. Strengthening patent protection

First, I simulate the effects of strengthening patent protection. In each case, I consider a decrease in the initial value of δ reported in [Table 1](#) that corresponds to a 25% reduction in the relative insecurity of patents and secrecy ($\delta - \ell$). The simulation results are displayed in [Table 2](#) below.

Observe that in both simulations with $\mu = 1$, the model reproduces the traditional growth enhancing effect of stronger patent protection. Since there is no change to patent utilization, there is no change to the effective cost of R&D for either value of α , and the increased reward from innovation generates a higher rate of economic growth. Although leaders' strengthened monopoly positions imply an increase in average mark-ups (η), this effect is partially offset by an increase in equilibrium consumption expenditure. Still, the policy change results in a decrease in total welfare from consumption. Thus, the policy produces the classic patent trade-off between growth and the welfare cost of increased monopoly power. As is common in endogenous growth models, since the initial equilibrium exhibits too little R&D from a social welfare

¹³ See [Appendix A](#) for a derivation of equilibrium in this case.

¹⁴ The value of δ in the $\mu = 1$ cases represent the weakest level of protection such that $\mu = 1$ is optimal. Necessarily, $\mu = 1$ remains optimal for all δ less than these initial values.

¹⁵ Alternatively, one could adjust κ to hold the cost of R&D constant across α for any fixed $\mu < 1$. This is equivalent to allowing β_R to adjust in the calibration, since β_R and κ influence the equilibrium only by entering the resource constraint as a $(\beta_R \kappa)$ term.

Table 2
Strengthening patent protection.

	Strong ($\alpha = 0.67$)				Weak ($\alpha = 0.33$)			
	$\mu = 0.45$		$\mu = 1.0$		$\mu = 0.45$		$\mu = 1.0$	
	Initial	$\delta \downarrow$	Initial	$\delta \downarrow$	Initial	$\delta \downarrow$	Initial	$\delta \downarrow$
$g(\%)$	0.500	0.378	0.500	0.547	0.500	0.377	0.500	0.522
μ	0.450	0.543	1.000	1.000	0.450	0.672	1.000	1.000
$\psi(\%)$	0.614	0.560	1.321	1.444	0.647	0.729	1.394	1.458
c	1.173	1.183	1.174	1.177	1.171	1.178	1.173	1.174
η	1.250	1.252	1.250	1.258	1.250	1.248	1.250	1.255
$L_{R\&D}(\%)$	3.666	3.047	3.591	3.928	3.789	3.116	3.726	3.897
$\frac{m(c,\mu)}{\rho-n}$	-1.195	-1.088	-1.332	-1.387	-1.202	-1.096	-1.318	-1.346
U	0.194	-0.037	0.057	0.131	0.188	-0.047	0.070	0.106

Table 2 displays results from a decrease in δ corresponding to a 25% decrease from each initial value of $(\delta - \ell)$. The final three rows report the share of labor employed in R&D, welfare from consumption discounted to time zero, and total discounted welfare as specified in Eqs. (4.2) and (4.3).

perspective, the welfare effect of the change in economic growth dominates this consumption effect and aggregate welfare increases.

However, in both cases where leaders use a mix of patents and secrecy in their appropriation strategies, this policy conclusion is reversed. In accordance with Proposition 2, the shift into patenting results in an increase in the effective cost of R&D that dominates the reward effect. Indeed, the corresponding decrease in growth is essentially equivalent across the strong and weak blocking effect cases. This is because the smaller initial difference in $(\delta - \ell)$ in the weak blocking effect case implies that leaders' choice of patent utilization is more sensitive to changes in the strength of patent protection. That is, the weak blocking effect case exhibits a greater shift into patents as a result of the policy change, which compensates for the smaller contribution of patents to the cost of R&D at any common level of μ . Note that this difference in the relative magnitude of changes to patent utilization and economic growth illustrates that the patent rate may increase or decrease as a result of strengthened patent protection, as specified in Proposition 2.

On the other hand, the fact that leaders rely on a mix of patents and secrecy when $\mu = 0.45$ does mitigate the price effect of stronger patent protection. Indeed, note that mark-ups actually decrease as a result of the policy change in the weak patent case. This is because patents continue to offer less protection from imitation relative to secrecy, and the large shift out of secrecy offsets the imitation reduction from stronger patents. As discussed in Section 4, this disclosure effect coupled with the increased resources allocated to consumption due to reduced innovation does translate into a modest increase in welfare from consumption. However, since both initial equilibria exhibit suboptimally low levels of innovation, overall welfare falls along with the rate of economic growth.¹⁶

5.2. Growth enhancing policies

Given that the market equilibrium produces too little R&D from a welfare perspective, I now turn to comparing the effect of policy interventions that stimulate innovation. Following a common approach in the endogenous growth literature, I first simulate the effects of an optimal R&D subsidy that equates equilibrium R&D to its optimal value implied by the social planner's problem. Next, I compare this subsidy policy against separate changes to patent and trade secrecy protection that generate an *equivalent increase in the growth rate*. This allows me to meaningfully compare the welfare implications among policies that share a common growth effect. Since qualitative results are similar across the weak and strong blocking effect cases, I focus on the strong case in the main text. Simulation results in the weak blocking effect case are presented in Appendix C.

For the initial values of δ and ℓ , it is optimal to provide a large R&D subsidy of $\sigma = 0.4139$ (Table 3).¹⁷ While the subsidy succeeds in increasing the growth rate by over 0.3%, it also induces a substantial shift into patenting as leaders respond to the increased threat of replacement. This increases the effective cost of innovation, and implies that a significant increase in the labor devoted to R&D is required to generate the increase in growth. Necessarily, this reallocates labor away from production, and consumption expenditure falls. Although the shift into patenting does reduce average prices through the disclosure effect, overall welfare from steady state consumption falls following the subsidy. Of course, the total welfare

¹⁶ As discussed in Section 4, this represents the typical case since the model tends towards suboptimally low innovation due to the combined presence of innovation deterrence through RPAs and imperfect appropriability. In the paper's supplementary material, I consider the effects of strengthening patent protection across a broad range of plausible parameters and show that this policy result continues to hold.

¹⁷ Although this subsidy is on the higher end of corresponding calculations from the literature, this follows from the model's tendency towards suboptimally low R&D. For example, Şener (2008) and Minniti et al. (2013) calculate optimal subsidies of about 15% and 22% respectively when considering an equilibrium with a 0.5% rate of growth and perfect appropriability. However, Segerstrom (2007) examines prices below their monopoly level and finds a range of optimal subsidies between 31.5%–72.5%, albeit in the context of a significantly different model.

Table 3
Growth enhancing policies ($\alpha = 0.67$).

	Initial	$\sigma^* = 0.4139$	$\delta \uparrow \text{to } 0.477$	$\ell \downarrow \text{to } 0.234$
$g(\%)$	0.500	0.808	0.808	0.808
μ	0.450	0.726	0.285	0.362
$\psi(\%)$	0.614	1.599	0.628	0.798
c	1.173	1.110	1.157	1.175
η	1.250	1.236	1.245	1.275
$L_{R\&D}(\%)$	3.666	7.655	4.884	5.368
$\frac{m(c,\mu)}{\rho-n}$	-1.195	-1.953	-1.428	-1.493
U	0.194	0.291	0.816	0.751

effect of the optimal subsidy is still positive since the beneficial effect of increased growth dominates the reduced welfare from consumption.

The next column demonstrates that a substantial *weakening* of patent protection (16.9% increase in δ) can induce the same positive growth effect while mitigating the welfare losses from reduced consumption. Weakening patent protection induces a shift away from patents into secrecy. Although overall appropriability decreases (negative reward effect), the associated reduction in the cost of R&D stimulates growth. Importantly, the same growth increase is achieved with substantially fewer additional labor resources devoted to R&D. Although consumption expenditure still falls, this decrease is relatively modest. In addition, despite the shift into secrecy, weakened patent protection implies that average prices still decrease. Altogether, weakening patent protection increases welfare by more than the increase associated with the optimal subsidy level.

The final column shows that the same growth effect can be achieved by a significant *strengthening* of trade secret protection (22% reduction in ℓ). In this case, stronger protection for secrecy induces a shift out of patenting, thereby reducing the effective cost of R&D. Moreover, the overall increase in appropriability generates a positive reward effect and stimulates growth. Although the patent rate increases with the rate of innovation, these findings are consistent with the empirical evidence that stronger legal protection of secrecy in the U.S. is associated with an initial decrease in patenting and increase in R&D (Png, 2017a; 2017b).¹⁸ On the other hand, the stronger overall protection from imitation increases average markups. While this implies that consumption expenditure increases, the net effect of increased appropriability is a decrease in welfare from consumption. Thus, the policy change produces the traditional trade-off from intellectual property protection of increased monopoly power in exchange for stimulating innovation.

6. Conclusion

In traditional endogenous growth models, it is typically assumed that private incentives to conduct R&D stem from the patent system. A natural consequence of this assumption is that stronger patents provide a greater reward for innovating, thereby stimulating R&D investment and economic growth. In contrast, empirical evidence shows that firms rely heavily on secrecy to appropriate returns to their innovations and increasingly use patents strategically to impede competing innovations. Strengthening the legal protection afforded to patent holders has been found to generate an increase in patenting without the expected gains in innovation or productivity, a finding referred to as the “patent puzzle.”

In this paper, I present a novel model of endogenous growth that is free of scale effects, analytically tractable, and consistent with this empirical evidence. The model allows successful innovators to choose a mix of patents and secrecy as their overall appropriation strategy. While secrecy provides superior protection from imitation, patents allow right holders to influence the effective cost of competitors' R&D efforts through complementary RPAs investments. By incorporating this endogenous appropriation decision, the model offers a new channel through which policy changes influence economic growth and aggregate welfare. Moreover, the model allows for a direct comparison of several different growth oriented policies, including subsidies to R&D investment and changes to intellectual property protection. In the case of strengthening patent protection, the impact of endogenous appropriation strategies is important enough to *reverse* qualitative predictions of the growth effect associated with the policy change. Overall, numerical simulations suggest that strengthening protections for trade secrets or weakening patent protection can be more effective growth enhancing policies than subsidizing R&D.

Since the model is stylized, quantitative simulation results should be viewed with appropriate caution. Nevertheless, these simulations demonstrate the type of insights that the model is capable of producing. In addition, the model offers several natural directions for future research to build on the approach developed in this paper. For example, future work could explore a more general framework for the relationship between patenting and RPAs. One option is to consider RPAs investments that influence not only competitor innovation, but imitation as well. Another is to allow firms to choose among types of RPAs that may differ in their reliance on patents and secrecy. In either case, these extensions may influence the complementarity or substitutability between patenting and RPAs investment, creating interesting implications for policy

¹⁸ For example, Png (2017a) finds that state level adoption of the Uniform Trade Secret Act was associated with fewer patents after one year, but that this effect diminishes over time.

analysis. Finally, one could introduce meaningful transitional dynamics into the model by relaxing the assumption that firms can adjust their patent portfolio instantaneously. Although this would substantially increase the complexity of the model, I view this type of dynamic policy analysis as an important step towards a more comprehensive understanding of the relationship between innovator appropriation strategies and economic growth.

Declaration of Competing Interest

None.

Appendix A

A1. Patent dominant equilibrium

The paper's main analysis assumes that the parameters determining the blocking effect of patents (α) and relative protection from imitation of patents and secrecy (δ and ℓ) are such that patents do not strictly dominate secrecy as an appropriation mechanism. That is, I restrict α , δ , and ℓ to ensure that leaders choose $0 < \mu < 1$ in equilibrium. In this section, I explore the model's equilibrium when these parameters instead imply that the optimal appropriation strategy is $\mu = 1$ and show that the model replicates (Dinopoulos and Syropoulos, 2007) in the special case of $\delta = 0$.

For simplicity, let us assume that $\delta < \ell$. In this case, patenting offers superior protection against imitation in addition to a positive blocking effect for any $0 < \alpha \leq 1$, so it is clear that $\mu = 1$ is always optimal. That is, the patenting condition no longer applies since Assumption 1 is violated, and it is replaced by the optimal $\mu = 1$. Observe from (2.3) that α has no effect on the cost of R&D in this case, since $D(t) = \kappa X(t)$ when $\mu = 1$. The model's equilibrium is determined by substituting $\mu = 1$ into the resource constraint (2.22) and the relative profitability condition (2.24). We have,

$$1 - s = \frac{c}{\lambda} [1 + (\lambda - 1)\delta] + \frac{\beta_{RKS}}{\beta_x} I, \quad (\text{A.1})$$

$$c = \frac{(1 - \sigma)\beta_{RKS}\lambda(\rho - n + 2I)}{\beta_x(\lambda - 1)(1 - \delta)}. \quad (\text{A.2})$$

Solving gives a closed form expression for the equilibrium rate of innovation,

$$I = \frac{\beta_x(\lambda - 1)(1 - \delta)(1 - s) - \beta_{RKS}(1 + (\lambda - 1)\delta)(1 - \sigma)(\rho - n)}{\beta_{RKS}[2(1 + (\lambda - 1)\delta)(1 - \sigma) + (\lambda - 1)(1 - \delta)]}, \quad (\text{A.3})$$

which matches the resource constraint of (2.25) used in the main text when $\mu = 1$. From (A.3), we see that $I > 0$ if and only if

$$\frac{(\lambda - 1)(1 - \delta)}{(\rho - n)(1 + (\lambda - 1)\delta)} > \frac{\beta_{RKS}(1 - \sigma)}{\beta_x(1 - s)}, \quad (\text{A.4})$$

which essentially ensures that the returns to R&D are sufficiently high relative to its cost when $\mu = 1$. In other words, (A.4) is a necessary and sufficient condition for (2.25) to cross the horizontal axis at $\mu > 1$ when graphed in (μ, I) space. Evaluating at $\sigma = 0$, straightforward calculation shows that $\partial I / \partial \delta < 0$. That is, when innovators rely fully on patents to appropriate returns from innovation, strengthening patent protection increases the rate of innovation. This is because the pro-innovation reward effect of stronger patents remains present, while there is no increase in the blocking effect of patenting since μ remains equal to one.

Finally, consider the special case of perfect patent protection. When $\delta = 0$, the optimal $\mu = 1$ implies that no imitation occurs for any value of ℓ and that R&D difficulty depends only on the leader's choice of RPAs with $D(t) = \kappa X(t)$. These two features correspond directly to the baseline assumptions in Dinopoulos and Syropoulos (2007). Indeed, taking into account differences in notation, Eq. (A.3) and condition (A.4) with $\delta = 0$ are identical to the (Dinopoulos and Syropoulos, 2007) counterparts in their expressions (33) and (34), which are used to derive the model's key results. Therefore, the model presented in this paper offers a generalization of the (Dinopoulos and Syropoulos, 2007) model by incorporating endogenous patenting decisions based on the trade-off between protection against imitation and subsequent innovation. In the special case of $\delta = 0$, this trade-off is removed, and the model reduces to Dinopoulos and Syropoulos (2007) where all innovations are fully protected by patents.

A2. Fully mobile labor

The baseline model partitioned the labor force into a fixed proportion of specialized labor employed only in RPAs and general purpose labor employed in either manufacturing or R&D. In this section, I instead suppose that all labor is fully mobile across these three activities. This implies a single type of labor and a common wage rate, which I normalize to one. The primary effect of this single wage rate is that the optimal choice of RPAs investment no longer depends on an

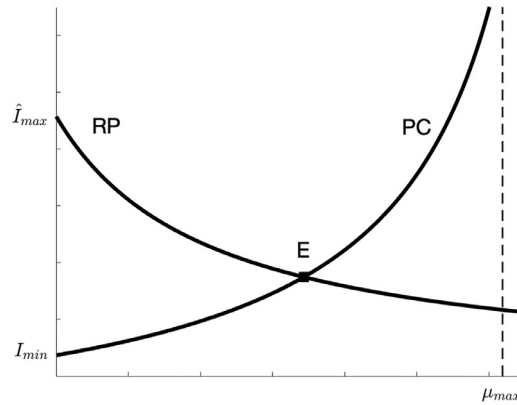


Fig. A.1. Equilibrium with fully mobile labor.

endogenously determined wage. Correspondingly, the ratio of the free-entry condition and the optimality condition for RPAs provides the following relative profitability condition that directly replaces Eq. (2.24),

$$I = \frac{\beta_x}{(1 - \sigma)\beta_R\kappa[\alpha\mu + (1 - \alpha)]}. \quad (\text{A.5})$$

As before, this condition describes the equilibrium rate of innovation in terms of the relative cost of R&D and RPAs. With mobile labor however, this relative cost is now entirely independent of c . The patenting condition of (2.20) remains unchanged since the wages do not directly enter either the cost or benefit of patents. This implies that the equilibrium values of I and μ are completely determined by the new relative profitability condition (RP) and the original patenting condition (2.20). Observe that Eq. (A.5) provides a downward sloping relationship between I and μ . This reflects the complementarity between patenting and RPAs: the increased effectiveness of RPAs implied by a larger μ implies an increase in the relative cost of R&D. Note that $I \rightarrow \beta_x/[(1 - \sigma)\beta_R\kappa(1 - \alpha)] \equiv \hat{I}_{\max}$ as $\mu \rightarrow 0$, and $I \rightarrow \beta_x/[(1 - \sigma)\beta_R\kappa]$ as $\mu \rightarrow 1$. Fig. A.1 illustrates the equilibrium by graphing these two equations in (μ, I) space. A unique equilibrium always exists provided Assumption 2 is adjusted to $\hat{I}_{\max} > I_{\min}$ to reflect the use of (A.5) rather than the RC as in the main text. Note that this version of Assumption 2 is still guaranteed to hold in the special case of $\alpha = 1$.

Since δ and ℓ do not enter Eq. (A.5), the impact of patent and trade secret protection now arise solely through their effect on the patenting condition. To understand why, note that any policy change that impacts $V(t)$ also impacts the incentive to use RPAs to extend the duration of market leadership. With a fixed proportion of specialized labor, the endogenous wage w_x adjusted the cost of RPAs to ensure that the market continued to clear. Without such cost adjustment, the use of RPAs scales with changes to $V(t)$ in exactly the same way that R&D does through the reward effect. In other words, any policy changes that generate a reward effect also generate an RPAs effect that offsets any impact to the equilibrium I .

It is straightforward to show that the model's core policy comparative statics continue to hold under the assumption of fully mobile labor. The main difference is that removing the reward effect eliminates the ambiguity in the relative magnitude of the change in μ and I , which determines the direction of change in the patent rate. We have the following,

Proposition 4. *In the model with fully mobile labor:*

- Subsidizing R&D expenditure increases firm patent utilization, increases the innovation rate, and increases the patent rate. That is, $\frac{\partial \mu}{\partial \sigma} > 0$, $\frac{\partial I}{\partial \sigma} > 0$, and $\frac{\partial \psi}{\partial \sigma} > 0$.
- Strengthening patent protection increases firm patent utilization, decreases the innovation rate, and increases the patent rate. That is, $\frac{\partial \mu}{\partial \delta} < 0$, $\frac{\partial I}{\partial \delta} > 0$, and $\frac{\partial \psi}{\partial \delta} \leq 0$.
- Strengthening trade secret protection decreases firm patent utilization, increases the innovation rate, and decreases the patent rate. That is, $\frac{\partial \mu}{\partial \ell} > 0$, $\frac{\partial I}{\partial \ell} < 0$, and $\frac{\partial \psi}{\partial \ell} \geq 0$.

Proof. All comparative static results for I and μ are apparent from shift analysis of the RP and PC curves in Fig. A.1. To show the patent rate ψ results, multiply both sides of the RP and PC curves by μ to obtain the following two conditions for the equilibrium ψ in terms of μ ,

$$\psi = \frac{\beta_x \mu}{(1 - \sigma)\beta_R\kappa[\alpha\mu + (1 - \alpha)]}. \quad (\text{A.6})$$

$$\psi = \frac{(\rho - n)(\delta - \ell)[\alpha\mu + (1 - \alpha)]\mu}{\alpha(1 - \ell) - 2(\delta - \ell)(1 - \alpha) - 3\alpha\mu(\delta - \ell)}. \quad (\text{A.7})$$

Note that (A.6) directly provides the equilibrium ψ in the special case of $\alpha = 1$. Otherwise, both (A.6) and (A.7) are upward sloping in (μ, ψ) space. The equilibrium ψ is given by their intersection at the equilibrium value of μ . Performing shift analysis exactly as before yields the comparative statics results. $\square$

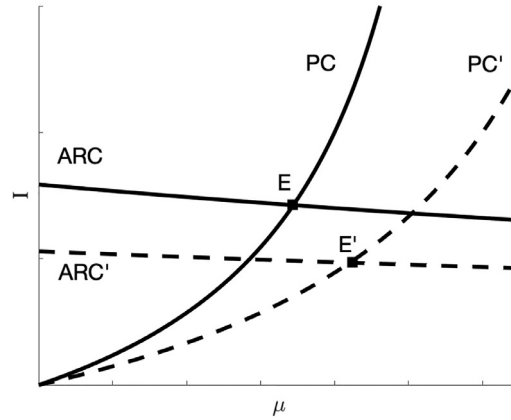


Fig. B1. Strengthening patent protection.

Appendix B

Proof of Proposition 2

The analysis included in the main text demonstrated that $\partial\mu/\partial\delta < 0$. It remains to be shown that $\partial I/\partial\delta > 0$ and that $\partial\psi/\partial\delta > 0$. First, rearrange the relative profitability condition (2.24),

$$\frac{c(\lambda - 1)(1 - \ell - \mu(\delta - \ell))}{\lambda[\rho - n + 2I]} = \frac{(1 - \sigma)\beta_R\kappa[\alpha\mu + (1 - \alpha)]s}{\beta_x}. \quad (\text{B.1})$$

Using the no-arbitrage condition (2.19) and the first order condition for patenting (2.18), this is equivalent to the statement

$$\frac{c}{\lambda} = I \frac{\alpha(1 - \sigma)\beta_R\kappa s}{(\delta - \ell)(\lambda - 1)\beta_x}. \quad (\text{B.2})$$

Eq. (B.2) provides an alternate version of the relative profitability condition that incorporates patenting behavior. As in the main text, this relative profitability condition can be plugged in to (2.22) to yield an alternate resource constraint (ARC),

$$I = \frac{(1 - s)\beta_x(\lambda - 1)(\delta - \ell)}{s\beta_R\kappa[\alpha(1 - \sigma)(1 + (\lambda - 1)\hat{\theta}(\mu)) + [\alpha\mu + (1 - \alpha)](\lambda - 1)(\delta - \ell)]} \quad (\text{B.3})$$

Just as in the main text, this can be graphed along side the patenting condition (2.20) to illustrate the model's equilibrium. In this case however, (B.3) captures both the reward effect, and the resource effect of the change in patenting behavior induced by a changed δ . As illustrated in Fig. B1, strengthening patent protection ($\delta \downarrow$) causes a leftward shift in (B.3) and necessarily results in a decrease in the equilibrium innovation rate.

Finally, note that the numerical analysis in the main text has shown directly that the patent rate may either increase or decrease following a decrease in δ .

Proof of Proposition 3

The analysis in the main text demonstrated $\partial I/\partial\ell < 0$. It remains to be shown that the impact of trade secret protection on firm patent utilization and the patent rate are ambiguous in general. Note that the numerical simulation in Section 5.2 provides an example in the “strong” blocking effect case where $\ell \downarrow$ implies $\mu \downarrow$ and $\psi \uparrow$. As shown in Appendix C, repeating the same policy analysis in the “weak” case provides a case when the magnitude of $\mu \downarrow$ is sufficiently large such that $\psi \downarrow$ following $\ell \downarrow$. To see the ambiguity in the direction of firm patent utilization directly, consider the special case of $\alpha = 1$. In this case, Assumption 2 is always satisfied and the resource constraint (2.25) and patenting condition (2.20) can be solved to provide the following expression that implicitly defines the equilibrium μ ,

$$\left[(2 - \sigma)(\delta - \ell)\kappa(\rho - n)\beta_{RS} \right] \mu^2 + \left[(1 - \sigma)((\lambda - 1)^{-1} + \ell)\kappa(\rho - n)\beta_{RS} + 3(1 - s)(\delta - \ell)\beta_x \right] \mu - (1 - s)(1 - \ell)\beta_x = 0 \quad (\text{B.4})$$

Application of the implicit function theorem shows that the sign of $\partial\mu/\partial\ell$ is the same as the sign of

$$(2 - \sigma)(\rho - n) \frac{\beta_{RKS}}{\beta_x} \mu^2 - (1 - \sigma)(\rho - n) \frac{\beta_{RKS}}{\beta_x} \mu + (1 - s)(3\mu - 1) \quad (\text{B.5})$$

Note (B.5) is negative as $\mu \rightarrow 0$ and positive as $\mu \rightarrow 1$. Further, since (B.5) does not directly depend on either δ or ℓ , it is straightforward to confirm that there exist feasible parameter values where δ and ℓ satisfy Assumption 1 and deliver $\partial\mu/\partial\ell > 0$.

Table C.1
Growth enhancing policies ($\alpha = 0.33$).

	Initial	$\sigma^* = 0.171$	$\delta \uparrow$ to 0.353	$\ell \downarrow$ to 0.292
$g(\%)$	0.500	0.593	0.593	0.593
μ	0.450	0.699	0.312	0.341
$\psi(\%)$	0.647	1.194	0.533	0.581
c	1.171	1.152	1.167	1.169
η	1.250	1.244	1.251	1.254
$L_{R\&D}(\%)$	3.789	4.948	4.246	4.297
$\frac{m(c, \mu)}{\rho - n}$	-1.202	-1.429	-1.295	-1.299
U	0.188	0.220	0.354	0.349

Welfare Calculations

Taking into account the four categories of prices discussed in [Section 4](#), sub-utility at time t can be written

$$\ln(u(t)) = \int_{\delta\ell}^{\delta} \ln(c(t)\lambda^{k(\omega,t)})d\omega + \int_{(1-\delta)\ell}^{(1-\delta)} \ln(c(t)\lambda^{k(\omega,t)-1}(1 + (\lambda - 1)(1 - \mu)))d\omega + \int_{\delta(1-\ell)}^{\delta} \ln(c(t)\lambda^{k(\omega,t)-1}(1 + (\lambda - 1)\mu))d\omega + \int_{(1-\delta)(1-\ell)}^{(1-\delta)} \ln(c(t)\lambda^{k(\omega,t)-1})d\omega \quad (\text{B.6})$$

$$= \ln(c) - \ln(\lambda) + \int_{\delta\ell}^{\delta} \ln(\lambda \cdot \lambda^{k(\omega,t)})d\omega + \int_{(1-\delta)\ell}^{(1-\delta)} \ln(\lambda^{k(\omega,t)}(1 + (\lambda - 1)(1 - \mu)))d\omega + \int_{\delta(1-\ell)}^{\delta} \ln(\lambda^{k(\omega,t)}(1 + (\lambda - 1)\mu))d\omega + \int_{(1-\delta)(1-\ell)}^{(1-\delta)} \ln(\lambda^{k(\omega,t)})d\omega \quad (\text{B.7})$$

$$= \int_0^1 \ln(\lambda)^{k(\omega,t)}d\omega + \ln(c/\lambda) + \left[\delta\ell \ln(\lambda) + (1 - \delta)\ell \ln(1 + (\lambda - 1)(1 - \mu)) + \delta(1 - \ell) \ln(1 + (\lambda - 1)\mu) \right] \quad (\text{B.8})$$

$$\ln(u(t)) = \ln(\lambda)It + m(c, \mu) \quad (\text{B.9})$$

Plugging this expression into [\(2.5\)](#), noting that c , I , and μ are constant in steady state yields the expression for welfare [\(4.3\)](#) used in the main text.

Next, to see that $m(c, \mu)$ can be non-monotonic in μ , note that

$$\frac{\partial m(c, \mu)}{\partial \mu} = \frac{\delta(1 - \ell)(\lambda - 1)}{1 + (\lambda - 1)\mu} - \frac{\ell(1 - \delta)(\lambda - 1)}{1 + (\lambda - 1)(1 - \mu)}. \quad (\text{B.10})$$

This is non-negative iff

$$\mu \leq \frac{(\delta - \ell) + \delta(1 - \ell)(\lambda - 1)}{(\lambda - 1)(\delta(1 - \ell) + (1 - \delta)\ell)}. \quad (\text{B.11})$$

The RHS is positive since $(\delta - \ell) > 0$ and $m(c, \mu)$ is increasing in μ when μ is low. However, [\(B.11\)](#) can fail as μ approaches 1 if either ℓ or λ are sufficiently large. Intuitively, the disclosure benefit of patenting decreases as ℓ increases and a larger λ implies a greater benefit to consumption smoothing across industries.

Appendix C

Growth enhancing policies - weak blocking effect

[Section 5.2](#) of the main text compared the welfare effects of the optimal R&D subsidy to changes in patent and trade secret strength that generate an equivalent increase in growth in the “strong” blocking effect case with $\alpha = 0.67$. In this section, I repeat this exercise in the weak blocking effect case with $\alpha = 0.33$. The associated calibrated values of δ , β_R , and λ are displayed in [Table 1](#). The results of the policy experiment are presented in [Table C.1](#).

Observe that the optimal subsidy level of $\sigma^* = 0.171$ is substantially lower than the value associated with the strong blocking effect analyzed in the main text. To understand why, recall the discussion of how calibrated parameters adjust to the weaker blocking effect in [Section 5](#). First, a weaker blocking effect requires a smaller calibrated value of δ in order to incentivize the targeted $\mu = 0.45$. This implies that innovations enjoy more overall protection against imitation, and that market R&D incentives increase relative to social incentives through the imperfect appropriation effect described in [Section 4](#). Similarly, the smaller calibrated values of β_R and λ imply both a lower total cost of R&D and a higher initial market rate of innovation to achieve the targeted 0.5% growth rate, since $g = \ln(\lambda)I$. Finally, as discussed in [Section 5.1](#), the smaller initial difference in $\delta - \ell$ implies that patent utilization is more sensitive to changes in conditions that impact the incentive to patent. Consequently, despite the much larger subsidy in the strong blocking effect case, the change in patent utilization associated with the optimal subsidy level is similar across cases. In other words, even though the increase in the cost of R&D from patenting is smaller per unit increase in μ when the blocking effect is weaker, the relatively larger shift into patenting generates a substantial resource wasting effect per unit increase in R&D subsidy.

Just as in the main text, intellectual property policy can be implemented to induce a shift out of patenting, reduce the cost of R&D, and match the subsidy's growth effect. Indeed, the increased sensitivity in patent utilization in the weak blocking effect case implies that this can be achieved with relatively small changes to δ and ℓ . In each case, changes to intellectual property policy generates the equivalent increase in economic growth with substantially fewer additional labor resources devoted to R&D. As before, this implies a smaller reduction in welfare from consumption, and an overall greater increase in total welfare.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.euroecorev.2020.103445](https://doi.org/10.1016/j.euroecorev.2020.103445).

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